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S-240 SURVEY AND SETTING OUT

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S.240 SURVEYING & SETTING OUT

1. SCOPE OF WORK

The scope of work under this section shall include furnishing labor, technicians, qualified professional surveyors, surveying equipment and materials required to perform all surveying work, setting out and calculations necessary to establish the accurate location of all plant facilities.

2. SERVICES PROVIDED BY THE OWNER

A basic survey on the field triangulation, polygon points and bench marks which has been performed by the Owner shall be taken as references only.

The Bidder shall obtain the plans containing the basic survey with all necessary information. Should the Bidder have any objection to this basic survey and the relevant drawings, he shall inform the Owner in writing within one week of this handling over by the Owner.

3. SERVICES TO BE PROVIDED BY THE BIDDER

3.1 General

The Bidder shall, as described in this section, perform all surveying and setting out and calculations necessary to establish the accurate location of all plant facilities and shall carry out a continuous survey during construction.

The Bidder shall set out, construct and establish bench marks in stable areas at the Site for his work. The survey for his bench mark grid shall only be based on approved triangulation points of the national grid. The Bidder shall submit calculations and drawings for the triangulation work done to establish the bench marks.

The Bidder shall submit in sufficient advance, for approval by the owner, the methods of surveying work.

3.2 Protection of Bench marks

The Bidder shall protect, preserve and keep accessible the bench marks of the basic survey and any provided by himself. Any damage or removal of bench marks shall be avoided as far as possible. Any accidental damage to any bench mark shall immediately be brought to the attention of the Owner. It is expressly

understood that the Bidder will be held responsible for damage to the bench marks and resulting consequences. The Bidder shall thus take preventive measures subject to the approval of the Owner in order to minimize the possibility of destroying bench marks. Bench marks must be of stainless steel or cast iron. The steel bolt shall be embedded in an absolutely stable concrete block of stable size subject to the approval of the Owner. Inscriptions shall be durable and clear legible. Intermediate points of the Bidder's triangulation survey used in establishing bench marks may be marked by steel pipes or other pegs, subject to the Owner's approval. Every new bench marks shall be double-checked, as far as possible in relation to points other those which served to establish the point in question.

New benchmark shall be supported by the pile, unless otherwise directed by the Owner.

3.3 Personnel and Equipment

The Bidder shall entrust the surveying only to such persons whose training and experience have sufficient qualification and knowledge to ensure proper fulfillment of the survey tasks assigned to them. If requested, the Bidder shall give the Owner any information relating to the survey. For the performance of the survey, the Bidder shall have sufficient number of reliable, accurate and well adjustment instrument by accompanying calibration certificate, the type and quality of which shall be subject to approval of the Owner.

3.4 Auxiliary Works

The auxiliary works of this section shall include, but not be limited to, the following:

- 1) Performance of all necessary survey calculations with clear presentation of calculations and results in order to facilitate verifications.
- 2) Exposing covered fix-points.
- 3) Removing machinery and obstructions from the required sightlines.
- 4) Stopping all machinery, drilling and blasting, pile driving and other work causing ground vibrations, stopping all smoke, dust, gas etc. hindering view causing refraction.
- 5) Restricting or stopping traffic of persons and vehicles near instruments or in sightlines during instrument observations.
- 6) Providing an adequate number of labors experienced as surveying aides and materials as deemed suitable by the Owner for the control and auxiliary surveys.

- 7) Removing all obstructive accumulation of water.
- 8) Taking all necessary safety precautions.
- 9) Furnishing and making materials required in connection with control surveys.
- 10) Providing fix points in lieu of or in addition to the Owner or Bidder's fix points in accordance with the instructions of the Owner.
- 11) Making all necessary topographic survey both onshore and offshore for the verification of lines and grades and quantity takeoff.

4. ACCURACY OF SURVEYING

Horizontal distances shall normally be measured with optic or electronic distance measuring devices. Chaining with metallic tape shall be restricted to the measurement of short distances (maximum 30m) and will be rejected for use in horizontal distance surveys for traverses.

Angles shall be measured by the theodolite.

Elevations shall be determined by differential horizontal leveling.

The existing bench marks shall be used during the construction.

5. VERIFICATION OF WORK

All relevant elements of survey and setting out may be checked by the Owner. The Bidder shall notify the Owner for this check sufficiently in advance of performing the work. In addition, the Bidder shall assist, if required, the Owner at any time in the performance of the verification work. The Bidder shall keep clear all marks and sightlines required for this purpose.

Checks will usually be made during work breaks, but in cases of urgency, the Bidder shall restrict or stop affected work without being entitled to extra payment. Any control by the Owner shall not relieve the Bidder of his full responsibility for the accuracy of the position and dimensions of the work and parts thereof.

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